

END-SEMESTER EXAMINATION, February-2025

Applied Computational Thinking (CSE 1401)

Programme: B.Tech(CSE [all specializations])
Full Marks: 60

Semester: 1st
Time: 3 Hours

9. (a) What is the chocolate cake analogy of ML? 2
- (b) Explain some sub modules of Scikit-Learn library. 2
- (c) Using the numpy library, create a 1-dimensional array with values from 0 to 11 (inclusive) using np.arange(). Then, reshape this array into a 2x6 matrix. Display the resulting 2D array. 2
- 10 (a) What is a local variable? Explain with an example. 2
- (b) What is a library? Explain with an example. 2
- (c) You are given a dictionary representing an address book, where the keys are 'names' and 'numbers', and the values are lists of names and corresponding phone numbers. Convert this dictionary into a pandas DataFrame, and display the resulting DataFrame. 2

End of Questions

Subject/Course Learning Outcome	*Taxonomy Level	*Ques. Nos.	Marks
Understand the importance of logical reasoning to decompose problems, analyze them, and develop efficient solutions.	L2,L3	1a,1c,4a	6
Develop a foundational understanding of programming by exploring core concepts and applying these to solve problems through algorithm design.	L3,L4	7a,7b,7c,8a,10a	10
Develop an understanding of control flow and problem-solving techniques to design effective solutions for computational challenges.	L4,L5,L6	1c,6a	4
Utilize Python libraries to analyze, visualize, and model data, using classification and clustering techniques to synthesize insights and evaluate patterns.	L1,L4,L5	9a,9b,9c,10b,10c	10
Utilize computational thinking with Python to analyze data, identify patterns, optimize solutions, and solve real-world problems across different domains.	L5,L6	2a,2b,2c,5a,5b,5c,8b,8c	16
Utilize core concepts of computational thinking, including automata theory, computational biology, and information theory, to define problems, design solutions, and conduct in-depth analysis.	L5,L6	3a,3b,3c,4b,4c,6b,6c	14

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

- Answer all questions. Each question carries equal mark.
1. (a) What are the steps of computational thinking? 2
 - (b) Write a Python program that searches for a number in an array and prints "Number Found" if the number is found. Use for..... else. 2

- (c) Write a Python program that prints numbers from 1 to 20. However, if the number is divisible by 5, it should skip that number using continue. The program should break if the number is greater than 15. 2

2. India is surrounded by Bangladesh, Nepal, Bhutan, Sri Lanka and Pakistan. The capitals of these countries are Dhaka, Kathmandu, Thimpu, Colombo and Karachi respectively. 2

- (a) Create a dictionary to store the capitals of the neighbouring countries. 2

- (b) Write a Python program which allows the user to input the name of a neighboring country. Display the capital of the entered country. 2

- (c) Add India in that dictionary. Also modify the program if the user enters a country that is not a neighbor of India, display an appropriate message. 2

3. (a) Given a list of items `fruits = ["apple", "banana", "apple", "orange", "banana", "apple"]`, write a Python program to count the frequency of each fruit and store it in a dictionary. 2

- (b) Write output of the following program: 2

```
set_1 = {1, 2, 3, 4, 4, 5}
set_1.add(6)
set_2 = {4, 5, 6, 7, 8}
union_set = set_1.union(set_2)
intersection_set = set_1 & set_2
diff_set_1 = set_1 - set_2
diff_set_2 = set_2.difference(set_1)
print(union_set)
print(intersection_set)
print(diff_set_1)
print(diff_set_2)
```

- (c) Find the Smallest Multiple of 7 Greater than N using while loop using Python program. 2

4. (a) Write a Python program that calculates the Body Mass Index (BMI) given a person's weight (in kg) and height (in meters) and classifies the result according to these ranges: 2

- $BMI < 18.5$: Underweight
- $18.5 \leq BMI < 24.9$: Normal weight

- $25 \leq BMI < 29.9$: Overweight
- $BMI \geq 30$: Obesity

Hint: Formula: $BMI = \text{weight}/(\text{height})^2$

- (b) Explain the following methods of text files: `Seek()`, `read()`, `fileno()`, `readable()`. 2

- (c) Write a Python code snippet to open a file for writing. 2

5. (a) Write a function in Python program, `prime(n)` that checks if a number `n` is prime. 2

- (b) What are the four collection data types of Python? 2

- (c) Write a Python program that removes all digits from a string. 2

6. (a) Write a Python program that check if one set is a subset or superset of another. 2

- (b) Write a Python program that remove all duplicate elements of a list. 2

- (c) Write a Python program that find the elements present in the first list but not in the second one. 2

7. (a) Write a function in Python program `longest_word(words)` that returns the longest word in a list of strings. Test the function with a list of words. 2

- (b) Create a Python program that checks if the temperature is either below 0°C (cold) or above 30°C (hot) and alerts the user accordingly. Use Boolean operators to determine the conditions. 2

- (c) Write a function in Python program `count_vowels(string)` that counts the number of vowels in a given string. Test the function with some strings. 2

8. (a) Explain the types of recursion functions in Python with examples. 2

- (b) Write a function in Python `sum_recursive(n)` that computes the sum of integers from 1 to `n` using recursion. 2

- (c) Write a Python program that checks for Armstrong numbers in a given range from 1 to `n`. An Armstrong number is a number that is equal to the sum of its digits each raised to the power of the number of digits. 2